

Intrinsic Valuation: DCF and Dividend Discount Models

Student Exercise Booklet

Part I — Discounted Cash Flow (DCF)

The DCF exercises follow the Session 3 sequence: FCFF/FCFE → discount rate → terminal value → enterprise/equity value → value per share → terminal-value consistency checks.

Exercise 1. FCFF Valuation and EV-to-Equity Bridge

A company forecasts the FCFF below. WACC = 8.5%; perpetual growth after Year 5 = 2.5%; debt = €180m; excess cash = €25m; diluted shares = 50m.

Year	1	2	3	4	5
FCFF (€m)	42	47	52	56	60

1. Calculate FCFF6 and terminal value at the end of Year 5.
2. Calculate the present value of explicit FCFF and terminal value.
3. Calculate enterprise value.
4. Bridge to equity value and calculate value per share.

Exercise 2. Build FCFF from Operating Forecasts

Year-0 revenue is €1,000m. Tax rate = 25%; D&A = 4.0% of revenue; CapEx = 5.5% of revenue; operating NWC = 12% of revenue. WACC = 9.0%; terminal growth = 2.5%.

	Y1	Y2	Y3	Y4	Y5
Revenue growth	8%	7%	6%	5%	4%
EBIT margin	14.0%	14.5%	15.0%	15.5%	16.0%

1. Forecast revenue, EBIT and NOPAT for Years 1–5.
2. Calculate D&A, CapEx and Δ NWC. Use Δ NWC = 12% $\times$ (Revenue_t – Revenue_{t-1}).
3. Calculate FCFF = NOPAT + D&A – CapEx – Δ NWC.
4. Calculate terminal value and enterprise value.

Exercise 3. FCFE Valuation with Net Borrowing

Cost of equity = 10.0%; perpetual growth after Year 5 = 3.0%; diluted shares = 40m.

€m	Y1	Y2	Y3	Y4	Y5
Net income	80	88	96	103	109
D&A	20	21	22	23	24
CapEx	30	32	34	35	36
Δ NWC	5	6	6	7	7
Net borrowing	8	5	2	0	0

1. Calculate FCFE for each year using FCFE = Net Income + D&A – CapEx – Δ NWC + Net Borrowing.
2. Calculate terminal equity value at the end of Year 5.
3. Calculate equity value directly.
4. Calculate intrinsic value per share. Do not subtract debt again.

Exercise 4. FCFF versus FCFE — Match Cash Flow and Discount Rate

A firm has the following three-year cash-flow forecasts. After Year 3, both FCFF and FCFE grow at 2.5% perpetually. WACC = 8.0%; cost of equity = 10.0%.

€m	Y1	Y2	Y3
FCFF	70	76	82
FCFE	42	46	50

1. Value the firm using FCFF and WACC.
2. Value common equity directly using FCFE and the cost of equity.
3. State why the two calculations use different discount rates and produce different valuation objects.
4. Identify the error if FCFF were discounted at the cost of equity.

Exercise 5. Build the First Stable-Year FCFF

At the end of Year 5, NOPAT is €150m. Stable growth from Year 6 onward is 3.0%, stable ROIC is 10.0%, and stable WACC is 8.0%.

1. Calculate Year-6 NOPAT.
2. Calculate the stable reinvestment rate using g/ROIC .
3. Calculate Year-6 reinvestment and FCFF.
4. Calculate terminal value at the end of Year 5.
5. Explain why mechanically growing Year-5 FCFF is not necessary when stable economics are built explicitly.

Exercise 6. Full DCF with Sustainable Terminal Economics

Forecast FCFF for Years 1–5 is shown below. WACC = 8.5%. Year-5 NOPAT is €170m. From Year 6 onward, stable growth = 2.5% and stable ROIC = 10.0%.

Year	1	2	3	4	5
FCFF (€m)	80	92	104	114	122

1. Calculate the present value of explicit FCFF.
2. Build Year-6 NOPAT, reinvestment rate, reinvestment and FCFF from the stable-state assumptions.
3. Calculate terminal value and its present value.
4. Calculate enterprise value.
5. Calculate the percentage of enterprise value represented by the present value of terminal value.

Exercise 7. Implied Terminal Multiple as a DCF Cross-Check

Use the DCF result from Exercise 6. Assume Year-5 EBITDA is €180m.

1. Calculate the implied terminal EV/EBITDA multiple = Terminal Value / Year-5 EBITDA.
2. Explain what the implied multiple tells you about the stable-growth DCF assumptions.
3. If a mature-company benchmark were materially below the implied multiple, identify which DCF assumptions you would investigate first.

Exercise 8. WACC × Growth Sensitivity with Reinvestment Consistency

Use the explicit FCFF from Exercise 6 and Year-5 NOPAT of €170m. Stable ROIC remains 10%. Recalculate enterprise value for each WACC and g combination below. For every g , first recalculate the stable reinvestment rate = g/ROIC and Year-6 FCFF.

	$g=2.0\%$	$g=2.5\%$	$g=3.0\%$
WACC=8.0%			
WACC=8.5%			
WACC=9.0%			

1. Complete the 3×3 enterprise-value sensitivity table.
2. Identify the highest and lowest values.
3. Explain why changing g must also change stable reinvestment when ROIC is held fixed.

Exercise 9. Enterprise Value to Common Equity Value

An FCFF valuation produces enterprise value of €2,400m. Debt = €500m; excess cash = €120m; non-controlling interests = €40m; non-operating investments = €35m; diluted shares = 100m.

1. Calculate common equity value using EV less senior/non-common claims plus non-operating assets.
2. Calculate value per diluted share.
3. Explain why debt is subtracted in an FCFF valuation but should not be subtracted again after an FCFE valuation.

Exercise 10. Integrated DCF Master Case

Year-0 revenue is €1,200m. Tax rate = 25%; D&A = 3.5% of revenue; CapEx = 5.0% of revenue; operating NWC = 10% of revenue. WACC = 8.5%. Debt = €350m; excess cash = €90m; diluted shares = 75m. In the stable period, g = 2.5% and ROIC = 11.0%.

	Y1	Y2	Y3	Y4	Y5
Revenue growth	9.0%	7.5%	6.0%	5.0%	4.0%
EBIT margin	15.0%	15.5%	16.0%	16.5%	17.0%

1. Forecast revenue, EBIT, NOPAT, D&A, CapEx, Δ NWC and FCFF for Years 1–5.
2. Build Year-6 FCFF from stable NOPAT growth and the stable reinvestment rate g /ROIC.
3. Calculate terminal value, present value of terminal value, and enterprise value.
4. Bridge to equity value and value per share.
5. Calculate the implied terminal EV/EBITDA multiple using Year-5 EBITDA.
6. Calculate the percentage of enterprise value represented by terminal value and identify the assumptions that deserve the strongest sensitivity testing.

Part II — Dividend Discount Models (DDM)

Formula Reference — Session 3

FCFF = NOPAT + D&A – CapEx – Δ Operating NWC

FCFF = NOPAT – Reinvestment

FCFE = Net Income + D&A – CapEx – Δ NWC + Net Borrowing

Enterprise Value = $\sum [FCFF_t / (1+WACC)^t] + TV_n / (1+WACC)^n$

Equity Value = $\sum [FCFE_t / (1+Ke)^t] + \text{Terminal Equity Value} / (1+Ke)^n$

$TV_n = FCFF_{(n+1)} / (WACC_{\text{stable}} - g)$

Growth = Reinvestment Rate $\times$ ROIC; Stable Reinvestment Rate = g /Stable ROIC

Implied terminal EV/EBITDA = Terminal Value / Terminal-Year EBITDA

DDM: $P_0 = \sum [D_t / (1+Ke)^t]$

Zero-growth DDM: $P_0 = D / Ke$

Gordon Growth Model: $P_0 = D_1 / (Ke - g)$

Sustainable dividend growth: $g = \text{Retention Ratio} \times ROE$

Retention Ratio = $1 - \text{Dividend Payout Ratio}$

Exercise 11. Zero-Growth Dividend Perpetuity

A mature utility has committed to maintaining a constant annual dividend of €2.40 per share indefinitely.

Investors require an 8.0% return on the company's equity. No dividend growth is expected.

1. Calculate the intrinsic value per share using the zero-growth DDM.
2. If the share trades at €27.50, calculate the percentage difference between intrinsic value and market price.

Exercise 12. Gordon Growth Model

Nordic Consumer SA has just paid a dividend (D_0) of €2.50 per share. Dividends are expected to grow at a constant 4.0% per year indefinitely. The required return on equity is 9.0%.

1. Calculate next year's expected dividend D_1 .
2. Calculate the intrinsic value P_0 using the Gordon Growth Model.
3. Calculate the dividend yield and expected capital-gain yield implied by the model.

Exercise 13. Market-Implied Constant Growth

A company's shares trade at €60.00. The company has just paid $D_0 = €2.40$ per share. Its cost of equity is estimated at 9.0%. Assume the market price can be represented by a constant-growth DDM.

1. Solve algebraically for the perpetual dividend growth rate g implied by the market price.
2. Verify that the implied growth rate satisfies the Gordon model condition $ke > g$.
3. Calculate D_1 using the implied growth rate and verify that $D_1 / (ke - g)$ reproduces the €60.00 price.

Exercise 14. Sustainable Growth and Intrinsic Value

Atlas Foods reports current EPS of €5.00 and pays out 40% of earnings as dividends. Its expected long-run ROE is 15.0%. Assume the payout ratio and ROE can be sustained indefinitely. The cost of equity is 10.5%.

1. Calculate the retention ratio.
2. Calculate the sustainable growth rate $g = \text{Retention Ratio} \times \text{ROE}$.
3. Calculate D_0 and D_1 .
4. Estimate the intrinsic value per share using the Gordon Growth Model.
5. Check whether the model's $k_e > g$ condition is satisfied.

Exercise 15. Required Payout Policy for a Target Growth Rate

Meridian PLC expects a stable long-run ROE of 15.0%. Management targets sustainable earnings and dividend growth of 6.0% per year. Current EPS is €4.00, and the cost of equity is 10.0%. Assume the firm is already in stable growth.

1. Calculate the retention ratio required to support 6.0% growth.
2. Calculate the corresponding payout ratio.
3. Calculate D_0 and D_1 implied by this payout policy.
4. Estimate intrinsic value per share.
5. Explain numerically why a higher target growth rate, holding ROE constant, requires a lower payout ratio.

Exercise 16. Two-Stage DDM

Helios SA has just paid $D_0 = €1.50$. Dividends are expected to grow by 12.0% annually for the next three years. From Year 4 onward, dividends are expected to grow at a stable 4.0% indefinitely. The cost of equity is 10.0% throughout.

1. Forecast D_1 , D_2 and D_3 .
2. Calculate D_4 , the first stable-growth dividend.
3. Calculate the terminal value P_3 at the end of Year 3.
4. Discount D_1 , D_2 , D_3 and P_3 to time 0.
5. Calculate the intrinsic value P_0 and the percentage of P_0 represented by the present value of terminal value.

Exercise 17. Varying Growth Rates Before Stability

Orion Consumer Group has just paid $D_0 = €1.80$. Expected dividend growth is 15% in Year 1, 12% in Year 2, 9% in Year 3, and 6% in Year 4. From Year 5 onward, dividends are expected to grow at 3.5% indefinitely. The cost of equity is 9.5% throughout.

1. Forecast D_1 through D_5 .
2. Calculate the terminal value P_4 using D_5 .
3. Calculate the present value of each explicit dividend and the terminal value.
4. Estimate P_0 .
5. Calculate what percentage of intrinsic value comes from the terminal value.

Exercise 18. Changing ROE, Payout and Growth

Nova Technologies has current $\text{EPS}_0 = €4.50$. For Years 1–4, expected ROE and payout ratios are: Year 1: ROE 18%, payout 30%; Year 2: ROE 16%, payout 35%; Year 3: ROE 14%, payout 40%; Year 4: ROE 12%, payout 50%. For each year, use $g_t = (1 - \text{payout}_t) \times \text{ROE}_t$ and $\text{EPSt} = \text{EPSt-1} \times (1 + g_t)$. From Year 5 onward, ROE is expected to stabilize at 10% and payout at 60%. The cost of equity is 10.5% during Years 1–4 and 9.0% in the stable period.

1. Calculate g_1 through g_4 and EPS_1 through EPS_4 .
2. Calculate D_1 through D_4 using each year's payout ratio.
3. Calculate stable growth from Year 5 onward.
4. Calculate EPS_5 , D_5 and terminal value P_4 .
5. Discount the explicit dividends and terminal value at 10.5% to obtain P_0 .
6. Calculate the proportion of P_0 attributable to terminal value.

Exercise 19. Multi-Stage DDM with a Changing Cost of Equity

Vector Media has just paid $D_0 = €1.20$. Dividend growth is expected to be 20% in Year 1, 20% in Year 2, 12% in Year 3 and 7% in Year 4. Stable growth of 3% begins in Year 5. The annual required returns are 12% for Year 1, 12% for Year 2, 11% for Year 3 and 10% for Year 4. Once stable growth begins, the long-run cost of equity is 9%.

1. Forecast D_1 through D_5 .
2. Calculate P_4 using the stable cost of equity and stable growth rate.
3. Construct the cumulative discount factor for each of Years 1–4.
4. Discount D_1 through D_4 and P_4 using the appropriate cumulative discount factors.
5. Calculate P_0 .

Exercise 20. Integrated ROE–Payout–DDM Valuation

Aurelia Group has $EPS_0 = €6.00$. During Years 1–3, ROE is expected to be 17% and payout 35%. In Year 4, ROE is expected to decline to 14% and payout rise to 45%. In Year 5, ROE is expected to be 12% and payout 55%. Use $gt = (1 - \text{payout}) \times ROE_t$ and $EPSt = EPSt-1 \times (1 + gt)$. From Year 6 onward, ROE stabilizes at 10% and payout at 60%. The cost of equity is 10.5% for Years 1–5 and 9.0% in stable growth.

1. Calculate growth, EPS and dividend for each of Years 1–5.
2. Calculate stable growth from Year 6 onward.
3. Calculate EPS_6 and D_6 .
4. Calculate terminal value P_5 .
5. Discount all dividends and terminal value to obtain P_0 .
6. Calculate the terminal-value share of total intrinsic value.

Exercise 21. Reverse Engineering Market Expectations

Stellar Brands trades at €48.00 per share. It has just paid $D_0 = €2.20$ and has an estimated cost of equity of 9.5%. Assume a constant-growth DDM is an appropriate representation of the market's expectations.

1. Solve for the perpetual growth rate implied by the market price.
2. Calculate D_1 at the implied growth rate.
3. Assume long-run ROE is 12%. Calculate the retention ratio required to support the implied growth rate.
4. Calculate the corresponding payout ratio.
5. Using the implied payout ratio and current $EPS = €4.50$, calculate the dividend that would be implied by the earnings/payout assumptions. Compare it with $D_0 = €2.20$ and comment on the consistency of the assumptions.

Exercise 22. Full Analyst Case: Transition to Stable Growth

Zenith Healthcare has $EPS_0 = €5.50$. Management expects the following operating economics: Years 1–2: ROE 18%, payout 30%; Year 3: ROE 15%, payout 40%; Year 4: ROE 12%, payout 50%. Use $gt = (1 - \text{payout}) \times ROE_t$ and $EPSt = EPSt-1 \times (1 + gt)$. From Year 5 onward, ROE is expected to stabilize at 10% and payout at 60%. The cost of equity is 10.0% during Years 1–4 and 8.5% in stable growth. The current market price is €78.00.

1. Calculate growth, EPS and dividend for Years 1–4.
2. Calculate the stable growth rate from Year 5 onward.
3. Calculate EPS_5 and D_5 .
4. Calculate terminal value P_4 .
5. Calculate intrinsic value P_0 .
6. Calculate the present value of the explicit dividends and the present value of terminal value separately.
7. Calculate the percentage of intrinsic value represented by terminal value.
8. Compare intrinsic value with the €78.00 market price by calculating $(P_0 - \text{Market Price}) / \text{Market Price}$.
9. Perform a valuation check using stable growth rates of 3.5% and 4.5%, keeping the stable cost of equity at 8.5%. Recalculate P_0 in each case.